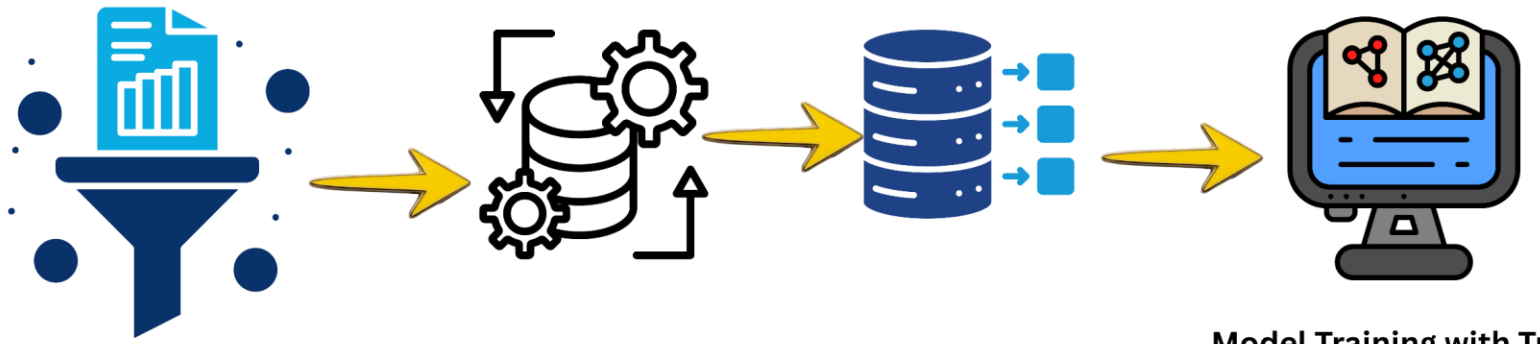


Data Collection
Using Kobo Collector

Data Preprocessing

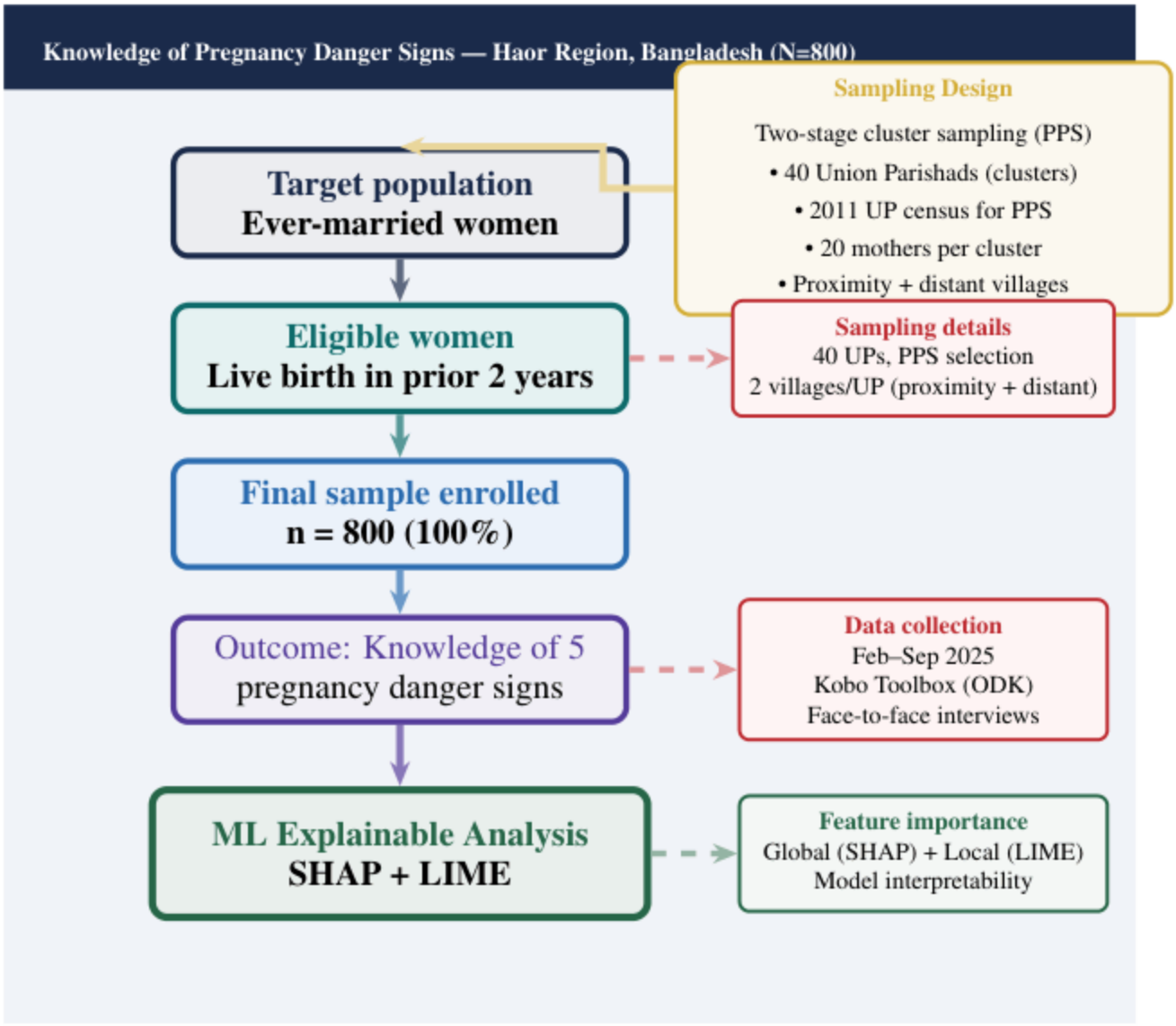
Boruta Algorithm

Model Training with Tuning

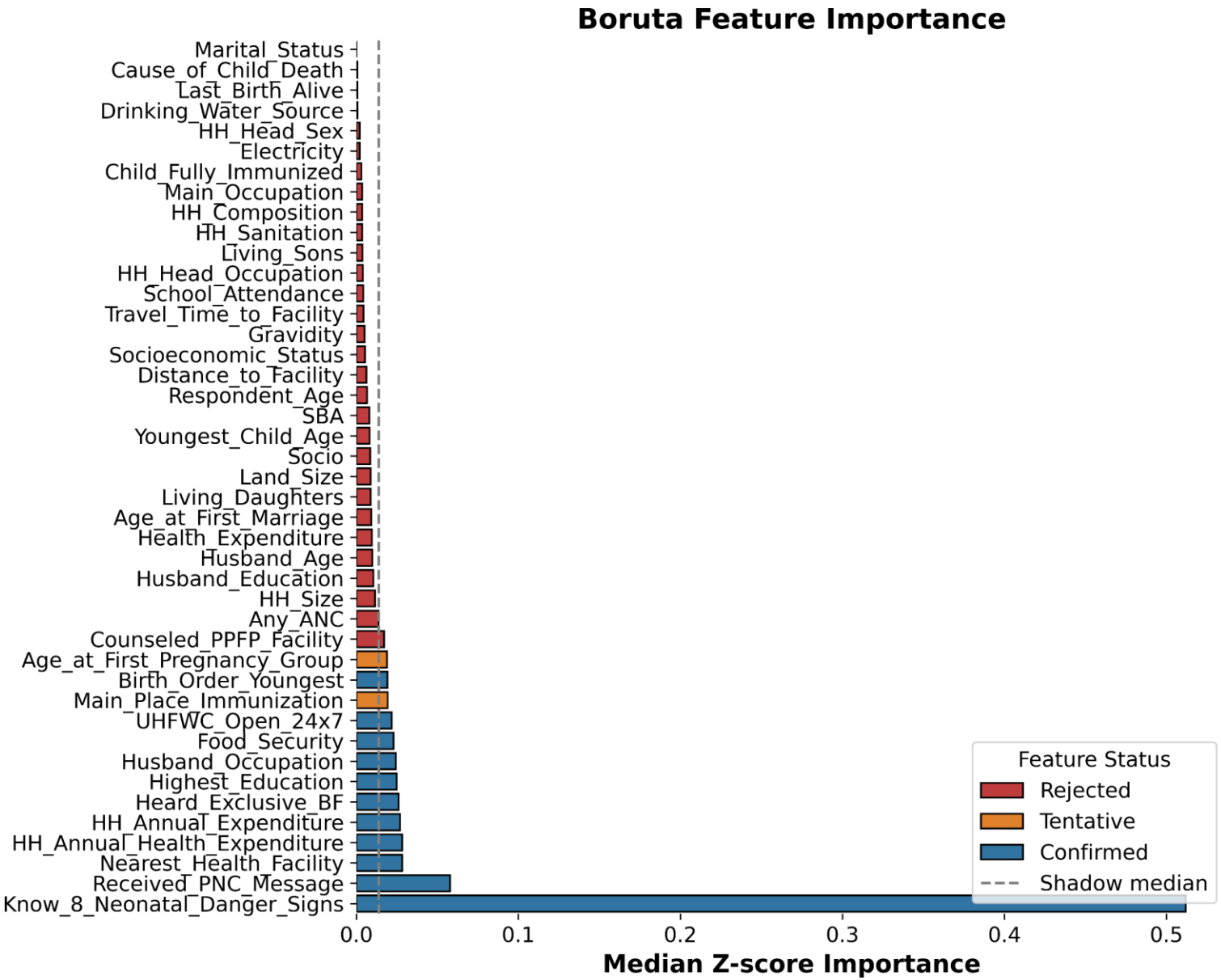




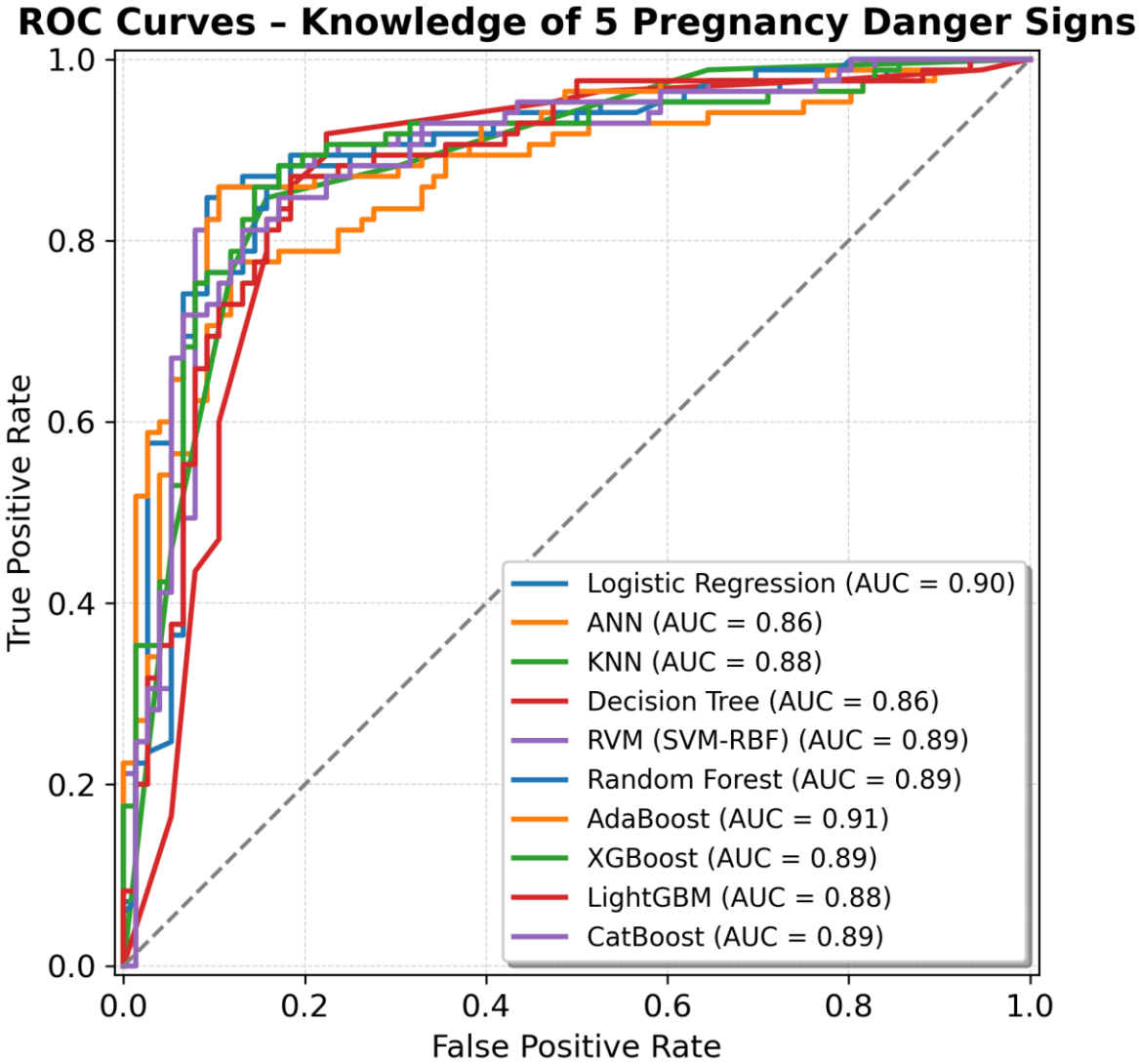
Explainable AI (SHAP and LIME)


Knowledge of Pregnancy Danger Signs — Haor Region, Bangladesh (N=800)
Sampling Design
Two-stage cluster sampling (PPS)
• 40 Union Parishads (clusters)
• 2011 UP census for PPS
• 20 mothers per cluster
• Proximity + distant villages
Target population
Ever-married women
Eligible women
Live birth in prior 2 years
Sampling details
40 UPs, PPS selection
2 villages/UP (proximity + distant)
Final sample enrolled
n = 800 (100%)
Outcome: Knowledge of 5
pregnancy danger signs
Data collection
Feb–Sep 2025
Kobo Toolbox (ODK)
Face-to-face interviews
ML Explainable Analysis
SHAP + LIME
Feature importance
Global (SHAP) + Local (LIME)
Model interpretability

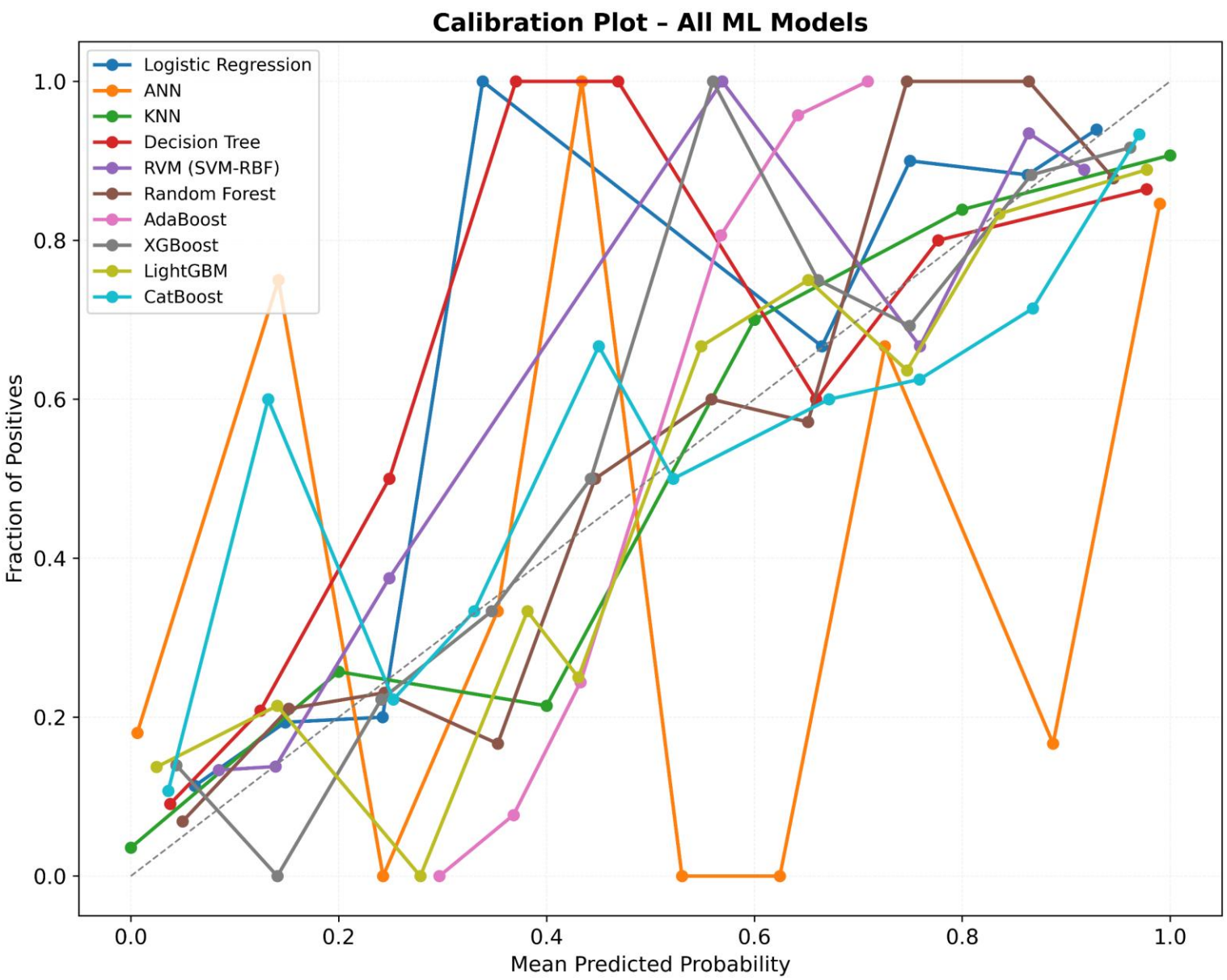
.


Boruta Feature Importance
Marital_Status
Cause_of_Child_Death
Last_Birth_Alive
Drinking_Water_Source
HH_Head_Sex
Electricity
Child_Fully_Immunized
Main_Occupation
HH_Composition
HH_Sanitation
Living_Sons
HH_Head_Occupation
School_Attendance
Travel_Time_to_Facility
Gravidity
Socioeconomic_Status
Distance_to_Facility
Respondent_Age
SBA
Youngest_Child_Age
Socio
Land_Size
Living_Daughters
Age_at_First_Marriage
Health_Expenditure
Husband_Age
Husband_Education
HH_Size
Any_ANC
Counseled_PPFP_Facility
Age_at_First_Pregnancy_Group
Birth_Order_Youngest
Main_Place_Immunization
UHFWC_Open_24x7
Food_Security
Husband_Occupation
Highest_Education
Heard_Exclusive_BF
HH_Annual_Expenditure
HH_Annual_Health_Expenditure
Nearest_Health_Facility
Received_PNC_Message
Know_8_Neonatal_Danger_Signs
0.0
0.1
0.2
0.3
0.4
0.5
Median Z-score Importance
Feature Status
Rejected
Tentative
Confirmed
Shadow median

.

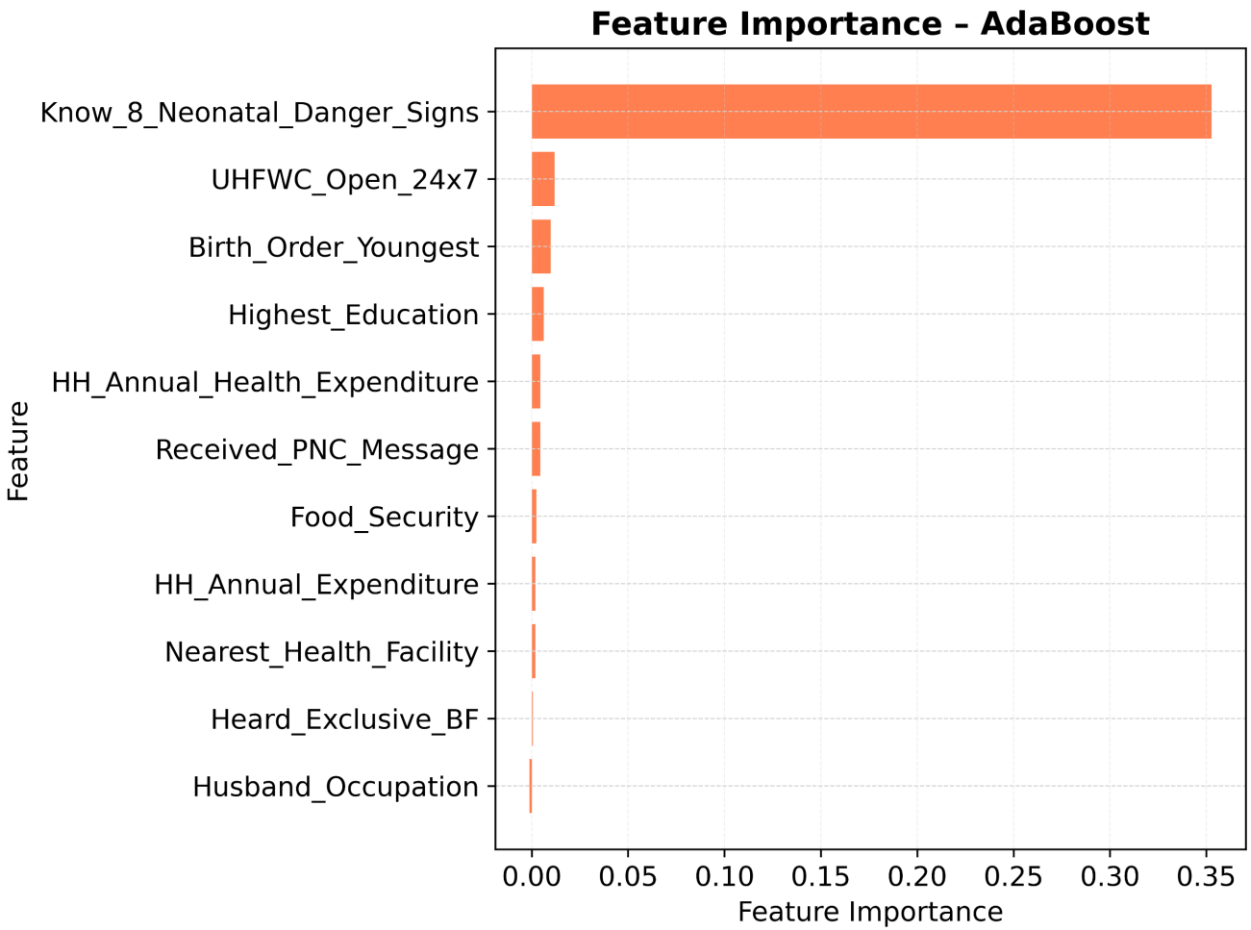

ROC Curves – Knowledge of 5 Pregnancy Danger Signs
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 0.90)
ANN (AUC = 0.86)
KNN (AUC = 0.88)
Decision Tree (AUC = 0.86)
RVM (SVM-RBF) (AUC = 0.89)
Random Forest (AUC = 0.89)
AdaBoost (AUC = 0.91)
XGBoost (AUC = 0.89)
LightGBM (AUC = 0.88)
CatBoost (AUC = 0.89)

.

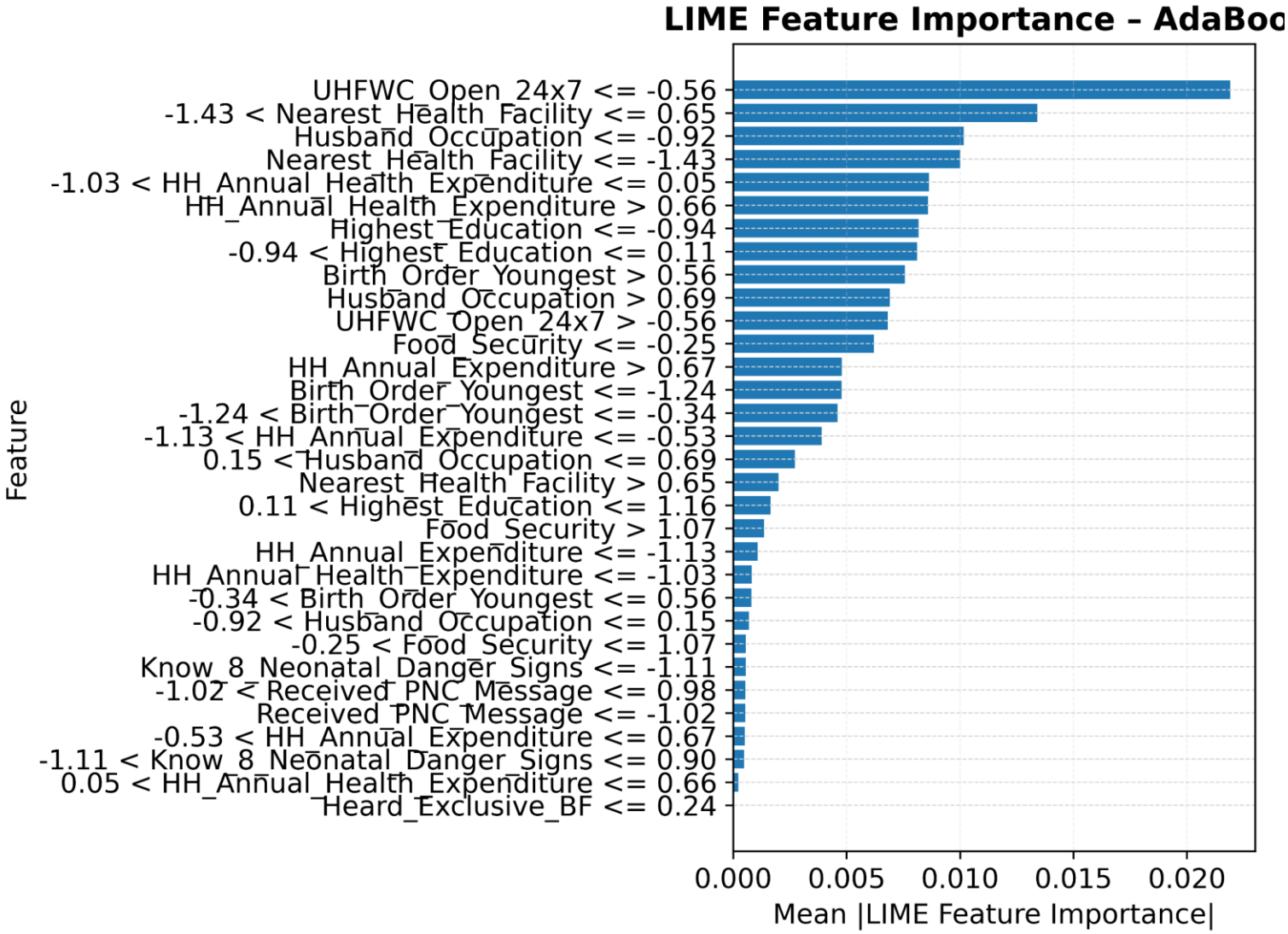

Calibration Plot – All ML Models
Logistic Regression
ANN
KNN
Decision Tree
RVM (SVM-RBF)
Random Forest
AdaBoost
XGBoost
LightGBM
CatBoost
Fraction of Positives
Mean Predicted Probability
0.0
0.2
0.4
0.6
0.8
1.0

.

...


Feature Importance – AdaBoost
Know_8_Neonatal_Danger_Signs
UHFWC_Open_24x7
Birth_Order_Youngest
Highest_Education
HH_Annual_Health_Expenditure
Received_PNC_Message
Food_Security
HH_Annual_Expenditure
Nearest_Health_Facility
Heard_Exclusive_BF
Husband_Occupation
Feature
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Feature Importance

...


LIME Feature Importance – AdaBoo
UHFWC_Open_24x7 <= -0.56
-1.43 < Nearest_Health_Facility <= 0.65
Husband_Occupation <= -0.92
Nearest_Health_Facility <= -1.43
-1.03 < HH_Annual_Health_Expenditure <= 0.05
HH_Annual_Health_Expenditure > 0.66
Highest_Education <= -0.94
-0.94 < Highest_Education <= 0.11
Birth_Order_Youngest > 0.56
Husband_Occupation > 0.69
UHFWC_Open_24x7 > -0.56
Food_Security <= -0.25
HH_Annual_Expenditure > 0.67
Birth_Order_Youngest <= -1.24
-1.24 < Birth_Order_Youngest <= -0.34
-1.13 < HH_Annual_Expenditure <= -0.53
0.15 < Husband_Occupation <= 0.69
Nearest_Health_Facility > 0.65
0.11 < Highest_Education <= 1.16
Food_Security > 1.07
HH_Annual_Expenditure <= -1.13
HH_Annual_Health_Expenditure <= -1.03
-0.34 < Birth_Order_Youngest <= 0.56
-0.92 < Husband_Occupation <= 0.15
-0.25 < Food_Security <= 1.07
Know_8_Neonatal_Danger_Signs <= -1.11
-1.02 < Received_PNC_Message <= 0.98
Received_PNC_Message <= -1.02
-0.53 < HH_Annual_Expenditure <= 0.67
-1.11 < Know_8_Neonatal_Danger_Signs <= 0.90
0.05 < HH_Annual_Health_Expenditure <= 0.66
Heard_Exclusive_BF <= 0.24
Feature
0.000
0.005
0.010
0.015
0.020
Mean |LIME Feature Importance|

.....